

A Practical Guide to Natural language processing

TOPIC -- NATURAL LANGUAGE PROCESSING

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General tendencies and (possible) future directions Based on long-standing trends in the field, it is possible to extrapolate future directions of NLP. As of 2020, three trends among the topics of the long-standing series of CoNLL Shared Tasks can be observed:

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A major drawback of statistical methods is that they require elaborate feature engineering. Since 2015, neural network-based methods have increasingly replaced traditional statistical approaches, using semantic networks and word embeddings to capture semantic properties of words. Intermediate tasks (e.g., part-of-speech tagging and dependency parsing) are not needed anymore. Neural machine translation, based on then-newly invented sequence-to-sequence transformations, made obsolete the intermediate steps, such as word alignment, previously necessary for statistical machine translation.

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Lexical semantics What is the computational meaning of individual words in context? Distributional semantics How can we learn semantic representations from data? Named entity recognition (NER) Given a stream of text, determine which items in the text map to proper names, such as people or places, and what the type of each such name is (e.g. person, location, organization). Although capitalization can aid in recognizing named entities in languages such as English, this information cannot aid in determining the type of named entity, and in any case, is often inaccurate or insufficient. For example, the first letter of a sentence is also capitalized, and named entities often span several words, only some of which are capitalized. Furthermore, many other languages in non-Western scripts (e.g. Chinese or Arabic) do not have any capitalization at all, and even languages with capitalization may not consistently use it to distinguish names. For example, German capitalizes all nouns, regardless of whether they are names, and French and Spanish do not capitalize names that serve as adjectives. This task is also referred to as token classification. Sentiment analysis (see also Multimodal sentiment analysis) Sentiment analysis involves identifying and classifying the emotional tone expressed in text. This technique involves analyzing text to determine whether the expressed sentiment

is positive, negative, or neutral. Models for sentiment classification typically utilize inputs such as word n-grams, Term Frequency-Inverse Document Frequency (TF-IDF) features, hand-generated features, or employ deep learning models designed to recognize both long-term and short-term dependencies in text sequences. The applications of sentiment analysis are diverse, extending to tasks such as categorizing customer reviews on various online platforms.

Terminology extraction The goal of terminology extraction is to automatically extract relevant terms from a given corpus.

Word-sense disambiguation (WSD) Many words have more than one meaning; we have to select the meaning which makes the most sense in context. For this problem, we are typically given a list of words and associated word senses, e.g. from a dictionary or an online resource such as WordNet.

Entity linking Many words—typically proper names—refer to named entities; here we have to select the entity (a famous individual, a location, a company, etc.) which is referred to in context.

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The premise of symbolic NLP is often illustrated using John Searle's Chinese room thought experiment: Given a collection of rules (e.g., a Chinese phrasebook, with questions and matching answers), the computer emulates natural language understanding (or other NLP tasks) by applying those rules to the data it confronts.

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Lemmatization The task of removing inflectional endings only and to return the base dictionary form of a word which is also known as a lemma. Lemmatization is another technique for reducing words to their normalized form. But in this case, the transformation actually uses a dictionary to map words to their actual form.

Morphological segmentation Separate words into individual morphemes and identify the class of the morphemes. The difficulty of this task depends greatly on the complexity of the morphology (i.e., the structure of words) of the language being considered. English has fairly simple morphology, especially inflectional morphology, and thus it is often possible to ignore this task entirely and simply model all possible forms of a word (e.g., "open, opens, opened, opening") as separate words. In languages such as Turkish or Meitei, a highly agglutinated Indian language, however, such an approach is not possible, as each dictionary entry has thousands of possible word forms.

Part-of-speech tagging Given a sentence, determine the part of speech (POS) for each word. Many words, especially common ones, can serve as multiple parts of speech. For example, "book" can be a noun ("the book on the table") or verb ("to book a flight"); "set" can be a noun, verb or adjective; and "out" can be any of at least

five different parts of speech. Stemming The process of reducing inflected (or sometimes derived) words to a base form (e.g., "close" will be the root for "closed", "closing", "close", "closer" etc.). Stemming yields similar results as lemmatization, but does so on grounds of rules, not a dictionary.

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when the amount of training data is insufficient to successfully apply machine learning methods, e.g., for the machine translation of low-resource languages such as provided by the Apertium system, for preprocessing in NLP pipelines, e.g., tokenization, or for post-processing and transforming the output of NLP pipelines, e.g., for knowledge extraction from syntactic parses.

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Where RMM is the relative measure of meaning token is any block of text, sentence, phrase or word N is the number of tokens being analyzed PMM is the probable measure of meaning based on a corpora d is the non zero location of the token along the sequence of N tokens PF is the probability function specific to a language Ties with cognitive linguistics are part of the historical heritage of NLP, but they have been less frequently addressed since the statistical turn during the 1990s. Nevertheless, approaches to develop cognitive models towards technically operationalizable frameworks have been pursued in the context of various frameworks, e.g., of cognitive grammar, functional grammar, construction grammar, computational psycholinguistics and cognitive neuroscience (e.g., ACT-R), however, with limited uptake in mainstream NLP (as measured by presence on major conferences of the ACL). More recently, ideas of cognitive NLP have been revived as an approach to achieve explainability, e.g., under the notion of "cognitive AI". Likewise, ideas of cognitive NLP are inherent to neural models multimodal NLP (although rarely made explicit) and developments in artificial intelligence, specifically tools and technologies using large language model approaches and new directions in artificial general intelligence based on the free energy principle by British neuroscientist and theoretician at University College London Karl J. Friston.